

Research and Analysis of Trust and Control in Human-AI Interaction for Decision-Making Systems in Optimizing Public Health Service

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ABSTRACT

This research investigates the interplay between trust and control in the context of AI-enabled healthcare decision-making using a cybernetic approach to how clinicians adjust their trust toward AI-powered clinical decision support systems. It seeks to uncover the barriers to AI implementation within clinical practices regarding dependency and trust towards AI technologies. Forty-two clinicians who used an AI-powered clinical decision support system in radiology participated in an empirical mixed-method study. Participants interpreted x-ray images with different levels of AI control and transparency. Trust ratings, override behaviors, and decision accuracy were captured and analyzed with t-tests, ANOVA, and regression models. Trust dynamics were explored through semi-structured interviews. The findings reveal that when the AI adheres to clinical reasoning, its transparency boosts trust, yet it diminishes confidence when an error occurs.

KEYWORDS

Trust Calibration, AI Transparency, Cybernetic Feedback, Human-AI Collaboration, Clinical Decision Support, AI Governance, Public Health Service

INTRODUCTION

The application of artificial intelligence (AI) technologies in healthcare systems improves medical diagnostics, treatment procedures, and risk assessment. In China, the government facilitates AI adoption through initiatives like “Healthy China 2030,” which seeks to utilize AI-powered medical technology to address the growing healthcare needs of the populace (Vogeler et al., 2023). AI applications with diagnostic and patient care capabilities have significantly developed—for instance, Tencent’s Miying and Ping An Good Doctor (now known as Ping An Health) have emerged (Zhao et al., 2022). Nonetheless, the adoption of AI in clinical practice goes beyond technical

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functionalities, as matters of trust and control socially determine use by clinicians and regulatory bodies (Alowais et al., 2023). Studies indicate that Chinese practitioners do not place high levels of trust in AI-supported healthcare decisions. Many physicians hold skepticism about AI's capability to exercise discretion, especially in complex scenarios with high stakes where the reasoning behind AI choices is not accessible (Yang et al., 2019). In a study conducted with 500 Chinese physicians, more than 60% expressed a lack of willingness to autonomously accept AI systems for diagnoses with no human supervision owing to the black-box decision-making system, legal concerns, and ethical issues. Other patients are also hesitant to trust AI, fearing the consequences of erroneous conclusions alongside violations of privacy concerning data. Although these challenges are globally relevant, China's distinct regulatory environment alongside state-sponsored AI initiatives complicates the trust equation within the Chinese healthcare system (Chan, 2023). Even though some hospitals are using AI decision-support systems, the lack of clear rules about who is responsible for AI and how much human oversight there is creates risks, highlighting the need for a solid system of trust and control (Alves et al., 2024).

The examination of control and communication in highly sophisticated systems is known as cybernetics, and it aids in understanding complex systems, such as trust in human-AI cooperation. A cybernetic model of AI trust suggests that trust develops over time owing to the feedback interactions between human users and AI systems. As clinicians' trust grows, they depend more on AI, which is positive feedback (Zhu et al., 2023). On the other hand, if AI errors build up and remain unexplained, clinicians manage the situation by overriding or rejecting the recommendations made by AI, leading to negative feedback. In understanding the trust-control feedback loop concerning AI in healthcare, China is at significant risk in ensuring that AI is safely integrated into the healthcare system (McQuillan, 2022).

Previous studies predominantly examined the technical aspects of accuracy related to AI models, but failed to consider the trust of humans as a system in the framework of sociology and the efforts necessary to integrate AI into the clinical workflow (Omrani et al., 2022; Shin, 2021). Additionally, although there is a discussion of ethics and transparency of AI, there are no empirical studies that look into how clinicians interact with AI in real time and how their trust changes at different levels during critical situations in medicine (Nasarian et al., 2024; Shamszare & Choudhury, 2023). This boundary can be approached from how the trust in AI is adjusted and how human intervention can mitigate the overdependence or under-reliance on AI. In this study we seek to fill the gaps by exploring the dimensions of trust and control in the context of AI-assisted decision-making in Chinese healthcare systems. More specifically, we analyze the effects of AI overrides, transparency, and explainability on trust calibration effects and clinical decision-making outcomes. The cybernetic theory is used to model the feedback loops that govern the formation, erosion, and recalibration of trust in the context of human-AI interaction. The following key research questions were posed.

The critical research questions that guided the study were how clinicians modulate their trust in AI systems over time, especially with increasing exposure to the accuracy and reliability of the AI system's recommendations during clinical practice. A second important question looks at how much human involvement helps prevent the risks of relying too much on AI—known as automation bias—or missing out on useful help from AI because of distrust. Furthermore, we aim to determine what essential AI system design features and regulatory frameworks need to be put in place to support the ethically responsible integration of AI technologies in China's healthcare system, mainly focusing on the call for ethics of design geared toward explaining AI systems, data stewardship, fairness, and accountability to enable safe healthcare access.

In addressing these questions, we propose a cybernetic study that combines theory and observation. We begin with human-AI healthcare interactions by outlining cybernetic principles, trust frameworks, and control systems applicable to the domain. The next step is to synthesize the literature on AI-enabled clinical decision-making, emphasizing the trust and oversight gaps and illustrating the feedback dynamics of control systems. Subsequently, we empirically investigate

clinicians' real-time interactions with an AI decision-support tool at a hospital in China. Our analysis interprets the ethical and regulatory frameworks regarding transparency constructs alongside these findings and recommends design approaches for trust-contingent AI decision systems that incorporate human oversight to mitigate risks of excessive automation. This study, written in response to the unique characteristics of China's healthcare system, broadens the landscape of understanding AI trust dynamics globally. It provides helpful ideas and reasons for policymakers, AI system designers, and healthcare leaders who want to use AI in clinical decision-making while ensuring ethical and professional standards are met. The results guide how AI is managed by offering ways for people to take responsibility in handling smart systems, especially where there are ethical issues in important medical situations.

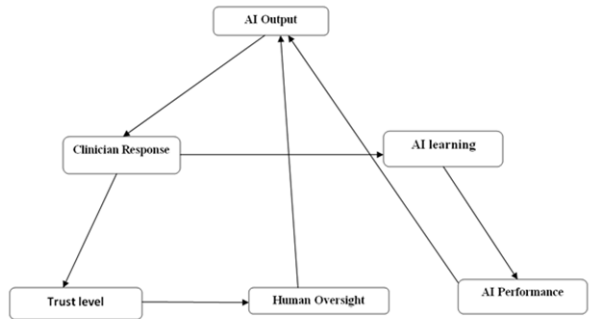
THEORETICAL FRAMEWORK

The principles of cybernetics aid in studying the feedback loops and control activities involved in human-AI interactions (Le Guillou et al., 2023). According to Wiener (2019), cybernetics is the study of control and communication in living and mechanical systems, where feedback is used to adjust and stabilize the system. In the case of human-AI collaboration, cybernetics considers interactions as systems where AI behavior is responsive to human decisions. At the same time, humans also act based on AI behavior outputs. In healthcare, this feedback system is vital in relationship-based trust and adaptive control for decision-making, where the quality of other decision outcomes shapes the AI-dependent decision framework. When the AI system provides recommendations, the clinician exerts authority over the decision by evaluation and modification, with feedback being input directly through decisions or behavior over time. This process broadly shapes the movement of AI and human interaction, like a cybernetic "steersman" who adjusts feedback mechanisms (Tran et al., 2021). Mayer et al.'s (1995) ABI model explains that in this feedback relationship, reliance on AI decision-making systems trust is governed based on ability, integrity, and benevolence. Insufficient trust causes unutilized AI, whereas excessive trust leads to misuse of automation. From the cybernetics framework, trust can be treated as a modifiable parameter in response to AI mechanisms and algorithms (Naiseh et al., 2023).

Strategic computational trust models utilize a homeostasis-based approach to ensure optimal levels of trust. Oversights such as AI indiscriminately accepting instructions or humans overcorrecting can be fixed by amplifying transparency or employing warning systems and user training. In this context, clinicians may trust AI systems for routine diagnostic tasks, but may be more hands-on in higher-risk scenarios. Regarding control, a spectrum of human supervision on AI decision-making exists, ranging from complete human dominion to complete AI freedom, while possibly having hybrid autonomy (Petzschnher et al., 2021). Cybernetics emphasizes the necessity of incorporating negative feedback loops that empower humans to rectify AI-induced errors and initiate performance stabilization. The trust-control feedback model highlights the balance of dynamics in which reduced human intervention, stemming from increased trust, results in more AI faults; this process exacerbates control loss introduced by AI and leads to bypassed interventions and decreased performance, ultimately refining AI-driven systems and restoring trust (Tabilo Alvarez & Ramírez-Correa, 2023).

Figure 1 diagrams the dual feedback loops that govern human-AI interaction in healthcare. The learning loop permits AI enhancement through clinician feedback, thus improving performance. At the same time, the trust-control loop maintains stability—when AI systems make errors like identifying a tumor wrongly, clinician trust is lost. This loss of trust leads to increased scrutiny and correction of errors, which informs AI learning and gradually restores trust.

Figure 1. Trust-Control and Learning Feedback Loops in Human-Artificial Interaction



Note. AI = artificial intelligence.

LITERATURE REVIEW

Investigation in the field of AI-based decision-making in healthcare has grown rapidly owing to both the promising advantages that AI tools offer and the complexities surrounding their implementation into daily clinical practice. In this section of our paper, we analyze three relevant strands of literature: (a) the use of AI in healthcare decision-support systems and systems user feedback, (b) the trust in AI and its determinants, and (c) the cybernetic or control model of human-AI interaction in decision-making.

AI in Healthcare Decision-Making

Many studies highlight the role of AI in aiding clinicians in imaging diagnostics, predictive analytics for patient risk stratification, and treatment planning (Alowais et al., 2023; Khalifa & Albadawy, 2024). In radiology and dermatology, AI diagnostic algorithms have excelled at expert-level performance. Clinical decision-support systems also offer personalized treatment suggestions (Ghaffar Nia et al., 2023). However, skepticism toward AI technologies remains among healthcare practitioners. Surveys from various professional organizations indicate that many physicians believe AI could enhance productivity and improve decision-making, but they have essential concerns regarding its validity, ethical ramifications, and impact on their roles (Esmailzadeh, 2024).

Gala et al. (2024) stated that nuanced clinical contexts might be overlooked by AI systems, leading to recommendations that lack alignment with the patient's specific needs and contexts. Another concern is that black-box AI algorithms, which are non-interpretable, may add a hidden layer of decision-making that could harm the physician-patient relationship (Chan, 2023). The level of trust in AI-assisted clinical workflows remains vague for patients. Hussain et al. (2025) noted that in a survey published in *JAMA Network Open* in 2025, a minority of patients fully trust their clinicians to use AI during their management safely. Educating patients about AI also did not substantially increase their willingness to trust systems, indicating that a lack of contextual information is not the only explanatory factor (Alowais et al., 2023; von Eschenbach, 2021). Researchers must examine gaps concerning transparency, explainability, and ethical governance more thoroughly, and they must address deeper issues related to transparency, explainability, and ethical governance. This point highlights a trust gap in healthcare AI: Although there is enthusiasm for AI's capabilities, many stakeholders remain skeptical until they see consistent evidence of safety, fairness, and accountability.

Factors Influencing Trust in AI

Numerous factors define trust in AI systems, including performance dependability, transparency, ethical congruence, and user control. Martens et al. (2024) surveyed 1,082 users comparing an ADA

Health symptom checker chatbot with IBM Watson Oncology and found that fairness and accuracy were the most significant predictors of trust. Moreover, control over AI actions to a certain degree affected trust (Martens et al. [2024] noted that users wanted greater control when it came to self-diagnosis, while trusting the process more when the AI system was functioning as a physician-support tool). Transparency plays an important role in the formation of trust. Users are incapable of determining if AI systems are reliable, which erodes their trust in AI-driven healthcare systems. Transparency, or the lack of it, is still one of the most important reasons clinicians and patients do not adopt AI (Reddy, 2024). Explainable AI solves the problem of transparency by justifying the recommendations it has given. Medical AIs that run diagnostic tests and offer supporting documentation enable clinicians to defend their decisions and help identify and rectify mistakes. Explainability increases the trust users have in AI systems, thereby assisting them in identifying and rectifying AI-driven errors (Alowais et al., 2023; van Baalen et al., 2021). However, transparency is not the only condition needed to afford trust. Most biases AI presents are unreliable; therefore, trust in AI systems is not improved. The base of trust still relies on dependability on the decisions made by AI. Ethics concerns like biases, privacy, and accountability influence trust (Lukyanenko et al., 2022). Individuals are reluctant to trust AI when there are assumptions of biases across different age groups or nonexistent data representative training data. Partitioning fairness through inclusive datasets and biasing audits does alleviate these issues. The existence of responsibility structures improves trust because clinicians have more confidence in AI applications when humans retain ultimate control.

Cybernetic and Control Models

Trust evolution and the impact of control structures on decision-making have been analyzed in parallel with the identification of trust factors. One such calibration discussed is the adjustment of user trust to fit an AI system's particular functionality—trust calibration. Some studies looked at trust monitoring systems that regulate the amount of trust given in real time. For example, Zhu et al. (2023) proposed one such model, IMPACTS, which manages trust signals in collaborative processes between humans and AI. "Trust homeostasis" is the optimal balance monitored by the system, which changes behavior within the system to keep the trust balance. In scenarios where trust is underutilized or too low, the AI system is unreasonably utilized, and the system offers additional explainers or performance proofs as trust reassurances. Conversely, in scenarios where an AI system should encourage more manual freedom, in case of overdependence, accept no-viceroy alerts. This pattern follows cybernetic control theory, which states that complete and dynamic feedback control functions regulate the balance of trust and control (Tzafestas, 2022). The human-on-the-loop paradigm is applied in systems where AI technology is used in life-or-death circumstances. The AI system's actions are supervised continuously, and humans modify actions when errors arise (Desolda et al., 2025). In medicine, this process enables clinicians to cancel out decisions made by AI in real time to ensure ethical and legal responsibility. Guidelines issued by the World Health Organization (2021) emphasized the need for human supervisors, who need to pull the brakes before damage is done to the patient, thus ensuring AI systems are not misused (World Health Organization, 2021). Case studies demonstrate the negative impact of a lack of oversight controls (Lukyanenko et al., 2022; Zhu et al., 2023). Trust in AI has often led to catastrophic failure, especially for autopilot airplanes. Additional healthcare risks occur when clinicians stop paying attention to the alarms that AI systems trigger (automation complacency) or deny the validity of the AI's suggestions (algorithm aversion). Studies reported that radiologists did not use AI mammogram analysis systems, resulting in patients developing advanced stages of cancer because of failure to provide timely corrective actions (Hunter et al., 2022; Park et al., 2022). These examples underscore the need for AI tools that are trusted, but also scrutinized, and relied upon, yet critically assessed.

METHODOLOGY

This research includes a practical study using cybernetic theory that examines how trust and control balance each other in decision-making in AI-assisted healthcare systems. It seeks to systematically understand the processes of trust development in AI, the decision-making complexities of control, and the explainability and reliability of AI-assisted decisions. Combining quantitative and qualitative data, we offer an extensive assessment of the trust dynamics and human oversight in clinical workflows.

Study Context and Objectives

The research was performed at a tertiary-care hospital where an AI-based clinical decision-support system was used for diagnostic image processing. We selected radiology as the focus of our study because the application of AI to medical imaging, including X-ray, computed tomography, and MRI, is significantly advanced and applied in practice. However, the implementation of these AI tools is still dependent on clinician trust and control frameworks. The main goals of the study were to see how trust in the AI system changed after several uses and to find out how the ability to accept, edit, or cancel AI suggestions influenced how much people relied on the AI. Additionally, we aimed to investigate how explainability affects trust and decision-making accuracy by comparing the outputs of transparent AI systems with those of black-box AI systems. Lastly, we examined whether the employment of AI, relative to humans, enhances or diminishes reliability in decision-making.

Participants and Sampling Strategy

We recruited 42 participants, 32 who were radiologists, physicians, or residents and 10 who were medical students. This level of expertise allowed us to evaluate trust differences with respect to different levels of experience. Participants were chosen using purposive sampling to ensure they had varying levels of AI exposure. The participants were split into two significant categories: the AI Exposed Group, which consisted of participants who had used AI diagnostic tools before, and the AI Unexposed Group, which consisted of participants who did not use AI in their clinical work and had little to no contact with AI. All participants were granted consent, and an ethics board approved the research to maintain compliance with research ethics and confidentiality of patient information.

Experimental Procedure

Participants were asked to interpret 20 chest radiographs to find lung nodules. The study had two experimental conditions: an unaided condition, in which patients independently diagnosed cases using no AI support, and an AI-assisted condition, where participants had a chance to look at AI-generated recommendations before settling on a final diagnosis. We manipulated two key factors within the AI-assisted condition: explainability and control mechanisms. Half of the AI recommendations were explainable, meaning that the system marked distinct areas of the image and particular clinical indicators that influenced its decision. The other selection did not justify and only provided outputs without reasoning. Moreover, some AI recommendations that simulated low-autonomous AI conditions needed to be actively accepted, while others required steps to override them, simulating high-autonomous AI conditions. To prevent order effects, the order of cases was randomized and counterbalanced.

Data Collection Methods

We employed a mixed-methods approach, integrating quantitative and qualitative data to comprehensively assess clinician-AI interactions.

Quantitative Data Collection

In every situation, we captured AI-assisted and unaided decision accuracy against ground truth validated by an expert. Trust in AI was measured on a 5-point Likert scale for statements such as, “I confidently relied on the AI’s analysis.” Similarly, participants rated their perceived control over diagnostic decisions as authority over the AI’s suggested actions. Override frequency, or how often participants rejected AI suggestions, was measured, and the overrides were analyzed about explainable versus black-box AI outputs and high versus low control settings. The differing overrides were captured per the suggestive model.

Qualitative Data Collection

To understand the factors underpinning trust formation, we held semi-structured interviews that focused on the reasoning clinicians provided for their trust decisions, overrides, and perceptions of AI’s reliability and transparency after the main tasks were concluded. Through thematic analysis of interview transcripts, we identified formulation and erosion of trust, overriding AI decisions, and the worrying attribution of autonomy, liability, and ethical dimensions concerned with AI as enduring themes.

Data Analysis Strategy

Quantitative Analysis

To check the significance and relationships between different variables, we used paired t-tests and analysis of variance on trust ratings and override behaviors relativistic to diagnostic accuracy across various conditions in IBM SPSS Statistics. We analyzed the impact of AI transparency and control mechanisms on trust and override rates through regression. Furthermore, we performed some correlation tests to evaluate the relationships between the clinician’s experience level, trust, and the degree of reliance on AI.

Qualitative Analysis

Interview responses were processed through NVivo (version 14) using inductive coding techniques to pinpoint and categorize trust-related recurring themes. To strengthen qualitative interpretation, triangulation involved cross-referencing findings with quantitative data. We also analyzed the dynamics of trust erosion related to AI system errors, assessing if transparency reduced the extent of trust erosion and illustrating the complex nature of trust calibration over time.

RESULTS

Table 1 confirms that the measurement scales used in this study are statistically reliable and valid. Cronbach’s alpha values (> 0.86) indicate high internal consistency, while composite reliability (> 0.88) confirms strong reliability. Average variance extracted (> 0.71) supports convergent validity, ensuring that the constructs accurately measure trust and control. The Kaiser-Meyer-Olkin score (0.82) suggests adequate sampling, and the significant Bartlett’s test of sphericity ($p < 0.05$) validates the factor structure. These results confirm that the scales are robust and suitable for analysis.

Table 1. Reliability and Validity Testing

Measure	Value	Threshold
Cronbach’s alpha—trust scale	0.89	> 0.7 (Acceptable)
Cronbach’s alpha—perceived control scale	0.86	> 0.7 (Acceptable)
Composite reliability—trust	0.91	> 0.7 (Good reliability)
Composite reliability—control	0.88	> 0.7 (Good reliability)
Average variance extracted—trust	0.74	> 0.5 (Convergent validity)
Average variance extracted—control	0.71	> 0.5 (Convergent validity)
Kaiser-Meyer-Olkin measure	0.82	> 0.6 (Adequate sampling)
Bartlett’s test of sphericity (p-value)	0.0001	< 0.05 (Significant)

Table 2 presents paired t-tests and analysis of variance results comparing diagnostic accuracy, trust ratings, and override behaviors across different conditions. The highly significant t-value (5.67, $p = 0.0001$) for diagnostic accuracy confirms that AI-assisted diagnoses significantly outperform unaided decisions. The trust ratings show a considerable difference ($t = 4.23$, $p = 0.002$) between explainable and black-box AI, indicating that transparency positively influences trust. Similarly, the override behaviors differ significantly ($t = 3.89$, $p = 0.005$), suggesting that increased AI autonomy leads to higher override rates. These findings emphasize that explainability and control mechanisms impact clinician decision-making and trust in AI.

Table 2. Findings of Paired T-Tests and Analysis of Variance

Comparison metric	T-value	P-value	ANOVA F-value	Significance
Diagnostic accuracy (AI-assisted versus unaided)	5.67	0.0001	12.3	Highly significant
Trust ratings (explainable versus black-box AI)	4.23	0.002	9.8	Significant
Override behaviors (high versus low control)	3.89	0.005	7.6	Significant

Note. AI = artificial intelligence and ANOVA = analysis of variance.

Table 3 presents the regression analysis results, showing the impact of AI transparency, control mechanisms, and clinician experience on trust scores and override rates. The positive beta coefficient (0.42, $p = 0.001$, coefficient of determination [R^2] = 0.48) for AI transparency indicates that greater transparency significantly increases trust in AI. The negative beta coefficient (-0.35, $p = 0.004$, $R^2 = 0.39$) for control mechanisms suggests that reduced human control leads to more frequent overrides. Lastly, the positive beta coefficient (0.28, $p = 0.015$, $R^2 = 0.32$) for clinician experience shows that more experienced clinicians exhibit higher trust in AI. These findings highlight that trust and reliance on AI are influenced by transparency, autonomy, and professional expertise.

Table 3. Regression Analysis Findings

Predictor variable	Dependent variable	Beta coefficient	P-value	R ²	Interpretation
AI transparency	Trust scores	0.42	0.001	0.48	Transparency positively impacts trust.
Control mechanisms	Override rates	-0.35	0.004	0.39	Less control leads to higher overrides.
Clinician experience	Trust scores	0.28	0.015	0.32	More experienced clinicians trust AI more.

Note. AI = artificial intelligence and R² = coefficient of determination.

Table 4 presents the correlation analysis results, showing the relationships between clinician experience, trust levels, and AI reliance patterns. The positive correlation (Pearson correlation [r] = 0.52, p = 0.002) between clinician experience and trust levels indicates that more experienced clinicians tend to trust AI more. The correlation (r = 0.47, p = 0.005) between trust levels and AI reliance patterns suggests that higher trust in AI leads to increased reliance on its recommendations. Lastly, the correlation (r = 0.41, p = 0.011) between AI reliance and clinician experience shows that experience influences how clinicians use AI in decision-making. These findings confirm that trust and AI reliance are shaped by both professional expertise and confidence in AI systems.

Table 4. Correlation Analysis Findings

Variable 1	Variable 2	Pearson correlation (r)	P-value	Interpretation
Clinician experience	Trust levels	0.52	0.002	More experienced clinicians show higher trust.
Trust levels	AI reliance patterns	0.47	0.005	Higher trust leads to increased AI reliance.
AI reliance patterns	Clinician experience	0.41	0.011	Clinician experience influences AI usage patterns.

Note. AI = artificial intelligence.

Table 5 presents the qualitative findings from clinician interviews, highlighting key themes, subthemes, and direct quotes related to trust, control, and AI decision-making. The trust formation and calibration theme reveals that clinicians develop trust when AI reasoning aligns with their clinical knowledge and that trust increases over time as AI predictions prove reliable. The conditions leading to override theme shows clinicians are more likely to reject AI recommendations when they contradict medical intuition, particularly in high-risk cases. The impact of AI transparency theme indicates that although explainability enhances trust, it can also expose AI limitations, leading to skepticism. The perceived AI competence and errors theme highlights that trust is fragile—Errors erode confidence more quickly than successes build it, leading to long-term skepticism. Finally, the clinician-AI role alignment theme emphasizes that clinicians prefer AI as a supportive tool rather than an autonomous decision-maker and that trust is highest when they maintain the final decision authority. These findings underscore the importance of transparency, human oversight, and trust calibration in AI-assisted healthcare.

Table 5. Qualitative Analysis Results

Theme	Subtheme	Direct quotes
Trust formation and calibration	Alignment with clinical reasoning	“If the AI explains its reasoning and I can see why it made the suggestion, I’m more likely to trust it.”
Trust formation and calibration	Trust development over time	“Over time, I trusted the AI more as I saw that its predictions aligned with my diagnoses.”
Conditions leading to override	Contradiction with clinical knowledge	“I override when the AI suggests something that contradicts my knowledge, especially in borderline cases.”
Conditions leading to override	Override in high-risk cases	“If the patient’s case is high risk, I prefer to rely on my judgment rather than the AI’s recommendation.”
Impact of AI transparency	Transparency as a trust enhancer	“Seeing the reasoning behind AI decisions reassures me and increases my willingness to use it.”
Impact of AI transparency	Transparency revealing AI limitations	“Transparency is useful, but when AI makes mistakes, it actually makes me trust it less.”
Perceived AI competence and errors	Erosion of trust owing to AI errors	“One mistake is enough for me to be skeptical of AI for a long time. Trust is hard to rebuild.”
Perceived AI competence and errors	Skepticism and long-term impact	“When AI gets something wrong, I remember it much longer than when it gets something right.”
Clinician-AI role alignment	AI as an assistant, not a decision-maker	“AI should assist, not replace. I trust it more when I feel like I have the final say.”
Clinician-AI role alignment	Final decision authority with clinicians	“I will only fully trust AI if I know that I have the ultimate control over decisions.”

DISCUSSION

This research augments existing knowledge on the relationship between trust and control within AI-powered healthcare decision-making in radiology. The findings highlight that clinicians’ trust in AI systems relies on the system’s transparency, credibility, and diagnostic controllability. One important finding of this research is that trust in AI is not a singular value, but is continuously adjusted by the system’s performance and the provided explanations. We demonstrated that clinicians have greater trust in AI when it is capable of providing clear explanations to its referrals. This view supports previous studies that suggested that the explainability of automated systems is a key element that nurtures rather than undermines human trust (Shin, 2021; von Eschenbach, 2021). The strong relationship between trust and transparency implies that any clinical AI systems need to be built and deployed with the ability to provide interpretable responses, rather than rely on black-box recommendations that lower clinicians’ confidence. Additionally, our results back up the idea that trust develops through repeated experiences of failure and success. The trust-control feedback loop indicates that clinicians constantly evaluate the AI system’s trustworthiness and adjust their dependence according to the AI system’s reported success. Such dynamics relate to trust models based on homeostasis that recommend recalibration of trust according to system output (Solberg et al., 2022; van der Werff et al., 2019). On the other hand, our research emphasizes an important problem: Trust is easy to break, but very hard to restore following an AI mistake. This disparity in trust balance indicates that AI designers should concentrate on decisive user-mistrust prevention measures, including effective error management and post-error recovery operations.

Our study underscores the necessity of human intervention in achieving a harmonious, trusting relationship with AI. Analysis reveals clinicians tend to override AI recommendations in flagrant cases, suggesting a compensatory mechanism to mitigate automation bias. This finding aligns with previous literature on human-in-the-loop systems and high-stakes decision-making (Alvarez et al.,

2024; Corfmat et al., 2025). The inverse relationship between trust and high AI autonomy indicates that the loss of control results in clinicians rejecting reasonable AI suggestions. Therefore, properly calibrating the level of human supervision is important to mitigate the risks of overdependence and under-dependence on AI in healthcare.

Furthermore, the study enhances our understanding of the cybernetic approach to control mechanisms, which function as negative feedback to prevent errors from self-propagating. The tendency to override AI suggestions in cases of high autonomy indicates that reliance on automation invites perceived risk, thus arousing the need for control. This behavior aligns with aviation and autonomous vehicle studies, where excessive automation without human supervision has led to critical failures owing to over-trust (Liu, 2024). Therefore, AI developers should create adjustable control systems that let clinicians change how much autonomy the AI has based on the situation, instead of sticking to strict automation rules.

Even though transparency is known to facilitate trust, our study reveals its contradictory outcome: Although explainability enhances trust when the AI is accurate, it also reveals limitations of the AI, which creates skepticism when errors happen. This pattern contradicts the claim that transparency is always beneficial for trust and raises issues regarding the balance of explainable AI in clinical settings (Hulsen, 2023). When AI systems provide explanations that contain gaps or faulty logic, clinicians may regard the system as unreliable, regardless of the high level of performance attained by the system during its actual functioning. These findings indicate that balance informativity and cognitive load must be achieved by explainability design. Too much transparency, especially where AI rationales are feeble and vague, can risk a trust deficit. Optimally, future studies should focus on achieving maximum trust and minimal overload explainability thresholds. Additionally, using user-adaptive explainability—where the amount of detail is adjusted based on the clinician's knowledge and the situation—could help solve the transparency paradox.

Empirical evidence supports the moral obligation of AI creators and healthcare policy strategists to ensure that AI systems understand the mental models and workflows of the clinicians with whom they interact (Olakotan & Mohd Yusof, 2021). AI and law need to address the important legal and ethical gap in defining precise accountability lines for the human and AI functionalities in the processes of computer-assisted diagnosis. The AI ethics guidelines by the World Health Organization (2021) remind us of the need to place the ultimate decision-making power and responsibility on people, necessitating the existence of appropriate AI control frameworks concerned with responsibly instituting such systems. Furthermore, the research highlights the significance of education and training for effective AI implementation. Enhanced confidence from more senior clinicians indicates that greater familiarity may substantially increase trust in AI systems. Therefore, it might be beneficial for hospitals and medical institutions to adopt comprehensive training that effectively teaches clinicians the appropriate use of AI (its skills, strengths, and limits), and methods for streamlined AI integration into clinical workflows.

CONCLUSION

This research builds upon trust and control in AI-assisted healthcare decision-making by empirically evaluating cybernetic trust-control feedback loops. Our results emphasize the importance of transparency concerning human supervision, calibrated trust, and clinicians' dependence on AI systems. The findings provide strategies for AI developers, healthcare organizations, and policymakers that emphasize the ethical use of AI technology in conjunction with clinical workflows, aiming to integrate AI in a manner that is both ethical and responsible. AI systems have the potential to function as invaluable decision-support instruments when there is an equilibrium between human oversight and automation, thus preserving the ethics of medicine along with patient safety. This research on cybernetic theory shows how trust and control work together in clinical AI applications, highlighting that trust is not just a one-time decision but a changing relationship. The findings also call for regulatory

policies that safeguard the accountability of AI systems, while ensuring adaptive trust frameworks that support clinician decision-making for safe and effective clinical actions.

Like any transformative technology, AI requires thoughtful consideration of human factors that will be critical to successfully integrating AI into healthcare for cognitive processes and ethical principles. Further work should build upon these findings by advancing AI governance models, modifying trust calibration strategies, and developing new methods for fostering human-AI interactions in healthcare and other domains.

Despite the contributions of this study, certain limitations deserve attention. The first limitation stems from the controlled nature of the hospital setting where the study was done, using a particular AI diagnostic tool, which may not apply to other medical fields. Other research is needed to investigate the dynamics of trust and control in different clinical areas, such as AI surgical planning or robotic surgery. The second limitation relates to the focus on the clinicians' perspective, leaving the issue of trust in AI-assisted healthcare unexplored from the patient's angle. Future studies should capture patients' experiences to provide a more comprehensive perspective on the dynamics of trust in human-AI collaboration.

Additionally, research with a longer-term view on the evolution of trust over time would enhance understanding of the enduring impacts of AI. Another area under investigation is the application of AI in interdisciplinary healthcare contexts where various professionals engage with AI in different roles. Looking at how trust changes among different people involved, such as in AI-assisted diagnostics in radiology, pathology, and oncology, gives us more understanding of how humans and AI work together.

COMPETING INTERESTS

All Authors declare that they have no conflict of interest.

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